

# ONKAR SARAMBALE

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## SUMMARY

Final-year Computer Engineering student (SGPA 8.9) with hands-on experience building end-to-end AI/ML pipelines, deploying NLP and computer vision models (BERT, ResNet-50), and developing LLM-powered applications using Vanna AI, Google Gemini, and LangChain. Built production-ready REST APIs (FastAPI, Flask), automated data workflows reducing manual effort by ~40%, and deployed models on Hugging Face Spaces. Seeking an AI/ML Engineering Internship to contribute to real-world GenAI and ML systems.

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, C++, SQL, HTML/CSS

**AI / ML:** PyTorch, TensorFlow, scikit-learn, BERT (fine-tuning), ResNet-50, Vanna AI 2.0, LangChain, RAG pipelines, NLP, Deep Learning, K-Means, Transfer Learning

**GenAI & LLMs:** Google Gemini (gemini-2.5-flash), Prompt Engineering, Hugging Face Transformers, LLM workflow automation

**Data & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Jupyter, FAISS, vector databases, Statistics

**Backend / APIs:** FastAPI, Flask, Node.js, Express.js, REST APIs, Pydantic, Swagger UI

**Cloud & Tools:** Hugging Face Spaces, Vercel, AWS (basics), Docker (basics), Git, MongoDB, SQLite

## PROJECTS

### DocMind-PDF-Q-A-Chatbot May 2026 – June 2026

Python · LangChain · FAISS · sentence-transformers · Streamlit

- Built a **Retrieval-Augmented Generation (RAG)** chatbot using LangChain + FAISS vector store for multi-document question answering with semantic search over uploaded PDFs
- Implemented document chunking, embedding generation with sentence-transformers, and **vector similarity search** for accurate context retrieval

### NL2SQL Clinic Chatbot Mar 2026 – Apr 2026

Python · Vanna AI 2.0 · FastAPI · Google Gemini (gemini-2.5-flash) · SQLite · Pydantic · Swagger UI

- Built an **end-to-end NL2SQL system** converting natural language queries to validated SQL using Vanna 2.0 + Google Gemini; served via FastAPI with full Swagger UI documentation
- Engineered a **SQL validation layer** blocking dangerous operations (DROP, DELETE, EXEC) before execution; pre-seeded DemoAgentMemory with 15 Q&A pairs boosting query accuracy
- Designed a **realistic clinic database** (5 tables, 200 patients, 15 doctors, 500 appointments) with skewed distributions and NULL injection to simulate real-world edge cases

### Indian Legal Document Classifier Feb 2026 – Mar 2026

Python · BERT (fine-tuned) · Flask · React.js · Hugging Face Spaces

- Fine-tuned BERT on a custom Indian legal dataset across 10+ document categories, achieving **92% validation accuracy** — demonstrating applied NLP and domain-specific transfer learning
- Designed an **end-to-end ML pipeline**: data preprocessing, tokenization, training, evaluation, and cloud deployment on Hugging Face Spaces
- Deployed full-stack application (Flask + React.js) enabling **real-time inference under 2 seconds**, serving as a scalable AI platform prototype

### SurplusShift – NGO Food Donation Platform Oct 2025 – Jan 2026

Python · PyTorch · ResNet-50 · Flask · React.js · Node.js · MongoDB · Computer Vision · GenAI

- Built a ResNet-50 food recognition model (PyTorch) achieving **92% accuracy** on Food-101 dataset (101 classes, 100K+ images) using transfer learning and data augmentation
- Designed a real-time image processing pipeline for automated food classification and donation tracking; integrated a **GenAI component** to estimate food quantities from images using LLM reasoning
- Implemented **GPS-based geo-validation** — fetching donor coordinates and cross-referencing with entered addresses to ensure accurate pickup routing (50m threshold)

### Mood-Based Song Recommendation System Oct 2024 – Nov 2024

Python · Pandas · scikit-learn · K-Means · Flask

- Developed a music recommendation engine using **K-Means clustering** on 10,000+ Spotify tracks, extracting audio features (valence, tempo, energy) for mood-aware predictions
- Built and deployed a real-time Flask REST API serving recommendations with **sub-second response time** after feature normalization and dataset cleaning

## INTERNSHIP EXPERIENCE

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### Data Analytics Intern — VOIS (Vodafone Intelligent Solutions) Oct 2024 – Nov 2024

- Automated data analysis workflows across 5+ business datasets using AI-LLMs, **reducing manual analysis time by ~40%** through structured AI-powered pipelines
- Implemented pipelines to extract actionable insights and generate structured reports using Python, Pandas, and LLM automation — improving reporting turnaround and decision quality
- Applied robust data cleaning and transformation techniques on real-world messy datasets, building reliable data engineering foundations

### MERN Stack Web Development Intern — EYGDS & AICTE Feb 2025 – Mar 2025

- Built and deployed a full-stack Recipe Sharing Platform (MongoDB, Express.js, React.js, Node.js) delivering 8+ REST API endpoints for auth, CRUD, and real-time data handling
- Reduced average page load time by ~30% through optimized React component structure and efficient MongoDB queries; followed MVC architecture and Git-based version control

## EDUCATION

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### Bachelor of Engineering – Computer Engineering Nov 2022 – May 2026

KJ College of Engineering and Management Research, Pune | **SGPA: 8.9**

## CERTIFICATIONS

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- Machine Learning Specialization — DeepLearning.AI, Stanford University (Feb 2026)
- Intro to Machine Learning — Kaggle (Jul 2025)
- Introduction to Generative AI — Google (Sep 2024)

## ADDITIONAL INFORMATION

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**Awards:** Techathon 2.0 by Innovation Foundation Pune (2025) — competed in applied AI problem-solving | **Research:** "SurplusShift: A Technology-Driven Approach for Efficient Surplus Food Redistribution and Food Waste Minimization," in the International Journal for Research in Applied Science & Engineering Technology (IJRASET), Volume 14, Issue V, May 2026. | **Languages:** English, Hindi, Marathi